

Hallucination Detection System

Evaluation Report

Based on 100 Questions · Models: GLM-4.7, GPT4.1 , LLAMA3.3:70B , Mistral

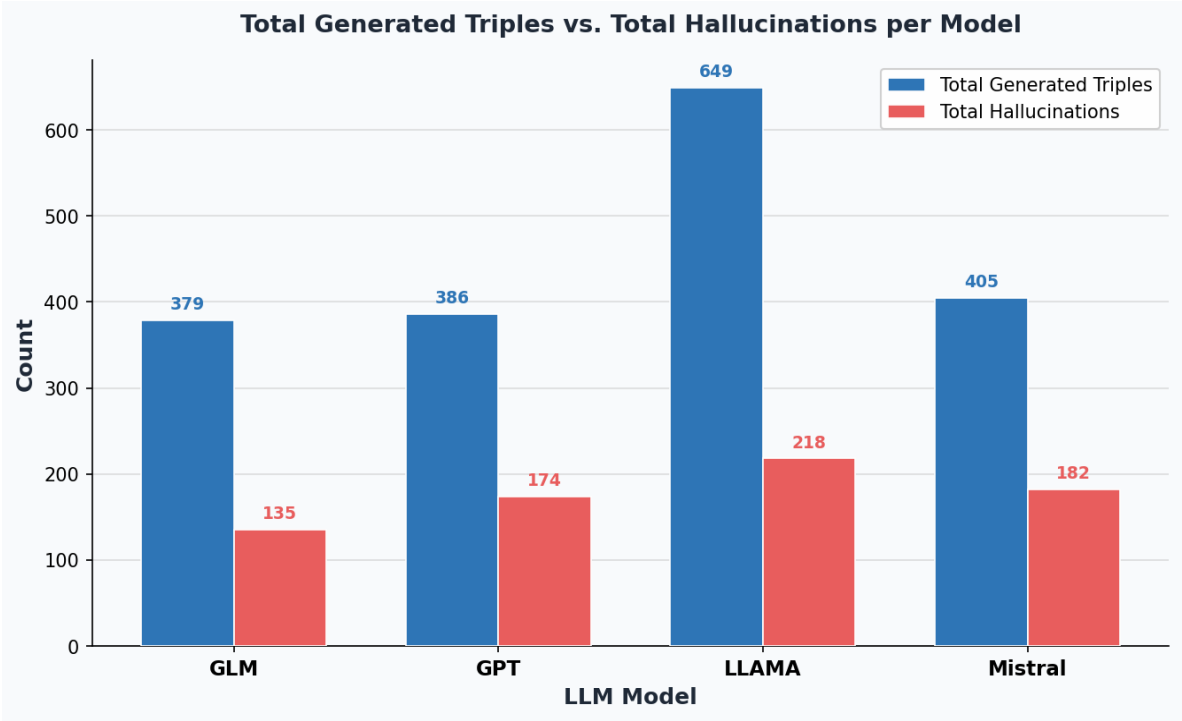
1. Hallucination Overview per Model

The table below shows the total triples generated, total hallucinations, and hallucination rate for each model.

Model	Total Generated Triples	Total Hallucinations	Hallucination Rate
GLM	379	135	35.62%
GPT	386	174	45.08%
LLAMA	649	218	33.59%
Mistral	405	182	44.94%
TOTAL	1819	709	38.98%

2. Generated Triples vs. Hallucinations — Visual Comparison

The grouped bar chart compares total triples generated (blue) against total hallucinations (red) per model.



3. System Accuracy — F1 Score Analysis

System accuracy is evaluated using the F1 score — the harmonic mean of Precision and Recall. $TP = \min(\text{detected}, \text{actual})$, $FP = \max(0, \text{detected} - \text{actual})$, $FN = \max(0, \text{actual} - \text{detected})$.
 $\text{Precision} = TP / (TP + FP)$ $\text{Recall} = TP / (TP + FN)$ $F1 = 2 \times (P \times R) / (P + R)$

Table 3a — Confusion Metrics per Model

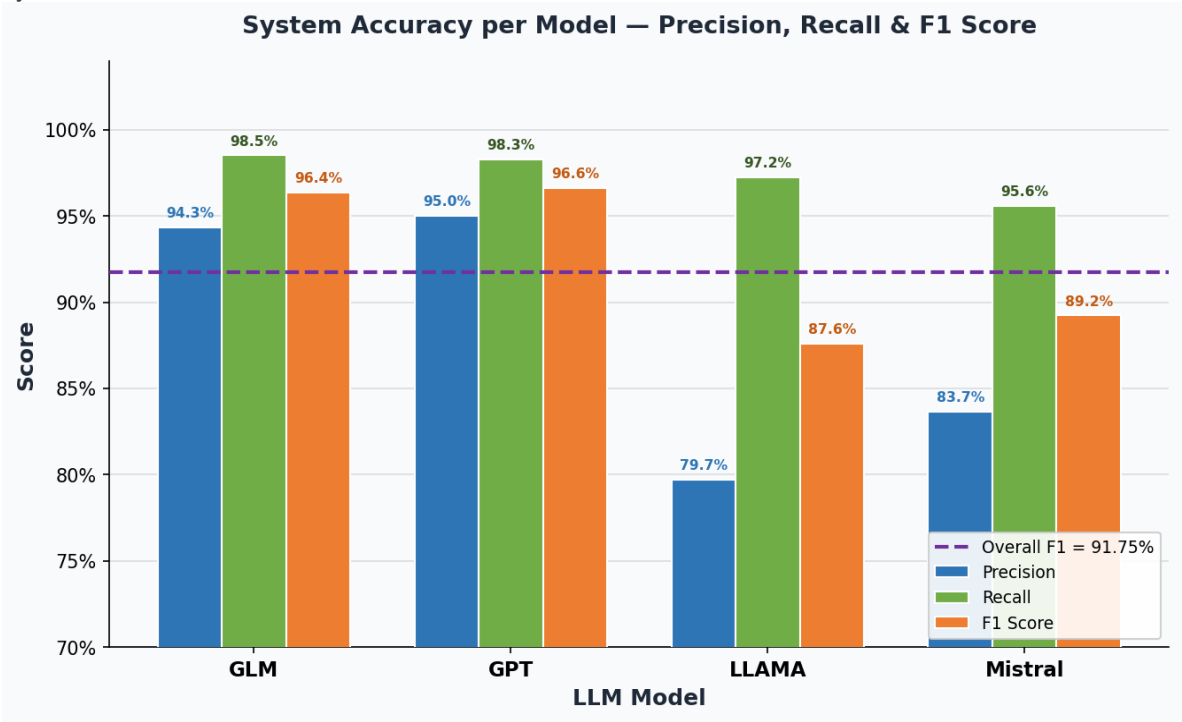
Model	TP	FP	FN	Precision	Recall	F1 Score
GLM	133	8	2	94.33%	98.52%	96.38%
GPT	171	9	3	95.00%	98.28%	96.61%
LLAMA	212	54	6	79.70%	97.25%	87.60%
Mistral	174	34	8	83.65%	95.60%	89.23%
OVERALL	690	105	19	86.79%	97.32%	91.75%

Table 3b — F1 Score Summary

Model	Precision	Recall	F1 Score
GLM	94.33%	98.52%	96.38%
GPT	95.00%	98.28%	96.61%
LLAMA	79.70%	97.25%	87.60%
Mistral	83.65%	95.60%	89.23%
SYSTEM TOTAL	86.79%	97.32%	91.75%

4. System Accuracy — Visual Comparison

Precision (blue), Recall (green), and F1 Score (orange) per model. The dashed purple line marks the overall system F1 of 91.75 %.



5. LLM alone Compare with LLM+System

Model	Hallucination without system	Caught by system TP	Missed FN	Detection rate
GLM	135	133	2	98.52%
GPT	174	171	3	98.28%
LLAMA	218	212	6	97.25%
Mistral	182	174	8	95.60%
OVERALL	709	690	19	97.32%

6. Types of hallucinations detected

Mistral

Hallucination Type	Generated	Detected Correctly	Not detected	Detected wrongly
Fabricated Fact	151	145	6	6
Rule violation from constraint	12	10	2	9
Predicate mismatch	12	12	0	0
Logical inconsistency	7	7	0	3
OVERALL	182	174	8	32

GPT

Hallucination Type	Generated	Detected Correctly	Not detected	Detected wrongly
Fabricated Fact	149	146	3	6
Rule violation from constraint	14	14	0	4
Predicate mismatch	12	12	0	0
Logical inconsistency	3	3	0	2
OVERALL	174	171	3	9

LLAMA

Hallucination Type	Generated	Detected Correctly	Not detected	Detected wrongly
Fabricated Fact	189	184	5	29
Rule violation from constraint	14	14	0	20
Predicate mismatch	4	3	1	4
Logical inconsistency	11	11	0	1
OVERALL	218	212	6	54

GLM

Hallucination Type	Generated	Detected Correctly	Not detected	Detected wrongly
Fabricated Fact	131	129	2	5
Rule violation from constraint	1	1	0	5
Predicate mismatch	0	0	0	0
Logical inconsistency	3	3	0	0
OVERALL	135	133	2	8

7. Key Insights

- The system is evaluated across 100 questions. 40 questions was focusing on questions involving single relationship from relationships in context , and 60 questions that cover 2 relationships with all possible combinations can be done from context.
- LLAMA is the Lowest hallucination rate with having largest number of hallucinations as llama tend in it's answers to add unverifiable info or additional info not asked for that is correct.
- The types of hallucinations generated is something that cannot be controlled from the LLM output so we did not have control of number of hallucination generated from each type but we tried to stress different types of hallucinations according to it's weight in the context , predicate mismatch is the lowest as there is only one relationship in the context that is presented for some and some not.
- The accuracy of the system is relatively high as the system is asking about closed domain and asking the LLMs for structured output so the process is easier, but with LLAMA the scenario is different as it sometimes provide additional info unstructured that cannot be detected and treated as unverifiable so it's F1 score is the lowest.
- The logical inconsistency violations is very small as LLM tends lower to hallucinate logically between different triples by providing single value for each subject and predicate and also if value is increasing or decreasing it keep the same sequence , same for tier violation.

8. Executive Summary — Key Performance Indicators

The table below consolidates all key performance metrics across models for at-a-glance comparison.

Metric	GLM	GPT	LLAMA	Mistral	System
Hallucination Rate	35.62%	45.08%	33.59%	44.94%	38.98%
Detection Rate	98.52%	98.28%	97.25%	95.60%	97.32%
Precision	94.33%	95.00%	79.70%	83.65%	86.79%
Recall	98.52%	98.28%	97.25%	95.60%	97.32%
F1 Score	96.38%	96.61%	87.60%	89.23%	91.75%